

Executive Summary

PASS RATE 34% target 90%	TOTAL TESTS 168 executed	PASSED 58 35% of total	FAILED 110 65% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 34% | Target: 90% | Gap: +56%

The Revenue Management System Kenya backend API testing reveals significant stability concerns with a 34% overall pass rate against the 90% target, indicating substantial issues with endpoint reliability and response correctness across both API automation and smoke validation suites. The consistent failure pattern across both test suites (33.75% and 35.23% pass rates respectively) suggests systemic backend issues rather than isolated endpoint problems, with approximately two-thirds of API endpoints exhibiting unexpected behavior, incorrect responses, or availability issues. The backend system requires immediate technical investigation to identify root causes of the widespread failures, as the current state presents risks to revenue processing operations and system reliability.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	27	49	36%	FAIL
2	BE Smoke Validation	88	31	57	35%	FAIL
TOTAL		168	58	110	34%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#43	02 Jul	168	34%	110	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	58	34.5%	180ms	Within SLA
Requests Failed	110	65.5%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	7%
Medium	4	4%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on the 106 failing endpoints to identify common failure patterns and systemic issues	Immediate	P1
2	Prioritize fixing critical revenue-impacting API endpoints that are part of core business workflows	3 days	P1
3	Review backend service logs and monitoring data to identify performance bottlenecks, timeout issues, or resource constraints	5 days	P2
4	Implement enhanced error handling and response validation in backend services to ensure consistent API behavior	1 week	P2
5	Establish incremental fix-and-verify cycles targeting groups of related endpoints to systematically improve pass rates	1 week	P2
6	Deploy additional backend observability tools to capture detailed diagnostics during API failures for faster resolution	2 weeks	P3

KEY FINDINGS

1	Backend API endpoints demonstrate 66% failure rate across 168 total test cases, indicating widespread service instability
2	Both API automation and smoke validation suites show similar failure patterns, suggesting consistent backend behavioral issues
3	53 API automation endpoints and 57 smoke validation endpoints are returning incorrect responses or experiencing failures
4	Current 34% pass rate falls 56 percentage points below the 90% target threshold for production readiness
5	Backend system reliability issues span across multiple API service layers based on comprehensive test coverage

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation



36%

